From Gibbs-Duhem to a local calculation

Guide page 1 | Use with worksheet B/C. Follow the worked C4 example, then attempt C5 independently.

1. Establish the path before using a residual

At fixed T,P, $x\, d\ln\gamma_1+(1-x)d\ln\gamma_2=0$. Differentiate $g=x\ln\gamma_1+(1-x)\ln\gamma_2$: the derivative is L plus that weighted derivative sum, so $dg/dx=L$. In the hand tables g means $g^E/(RT)$, not total mixing Gibbs energy.

On a general path, $\Sigma x_i d\ln\gamma_i=-h^E dT/(RT^2)+v^E dP/(RT)$. Isothermal VLE can have varying pressure; the teaching analysis neglects the excess-volume pressure term. A real isobaric local assessment needs independent h^E. A missing correction is not a failed measurement.

2. The g-first calculation used in C4/C5

Step 1: calculate g at all three compositions, with each row's own x weight. **Step 2:** calculate L at the middle point. **Step 3:** on these equally spaced grids, slope$\approx(g_{right}-g_{left})/(x_{right}-x_{left})$. **Step 4:** subtract center L to obtain r.

C4 value	Clean	Changed upper $\ln\gamma_1=0.36$
g(0.40)	0.48	0.48
g(0.50)	0.50	0.50
g(0.60)	0.48	0.504
L(0.50)	0	0
Slope, then r	0; 0	0.120; +0.120

C4's changed $g(0.60)=0.60\times0.36+0.40\times0.72$. Keep all other values fixed. **Now close the worked table and do C5:** use its new numbers; do not copy C4's answer.

3. What the calculation does not establish

A single finite-difference r is not a dataset-average score. Noise, rounding and spacing affect derivatives. The simple centered formula above assumes uniform spacing; Lab 03 uses a three-point Lagrange derivative with the actual spacing. Three interior points cannot establish a full-range area test.

The clean hand tables come from smooth consistent model functions; the changed entries are deliberately perturbed synthetic values. A nonzero numerical residual indicates a discrepancy under the stated operator and assumptions. It does not uniquely identify an experimental cause.

4. Keep scope labels literal

Not applicable: wrong path for the method. **Not assessed:** missing required information. **Meets/outside this criterion:** a particular eligible numerical test. Avoid collapsing these into a single overall pass/fail label.

For C1/C2, retain signed trapezoids. The Lab's normalized D uses positive and negative area magnitudes and may be undefined near ideality; it is not signed S.

Lab 03: reproduce the demo, then test predictions

Guide page 2 | Worksheet D1-D5, 65-95 min. Watch the real baseline demo, then operate the Lab yourself.

Open Lab 03: Consistency tests

D1. Baseline controls and two views

Set **Data set → Synthetic · isothermal, 350 K**, **Controlled perturbation → Consistent reference**, and **Sampling resolution → 41 points**. The default is 81, so change it explicitly. The synthetic h^E setting has no temperature contribution on this isothermal path.

Under **Inspect**, view **Log-activity ratio & integrated area**, then **Local Gibbs-Duhem residual**. Record D, local score, both statuses and the chart pattern. Expand **Inspect data & numerical results** if more digits are needed. Save the reference; wait until D5 for the fit.

D2-D3. Predict before the perturbation

Write both predictions on worksheet page 3 before selecting either defect. Then change **Controlled perturbation → Canceling errors · zero net area**; set **Amplitude=0.25**. Keep 41 points and the dataset unchanged. Results update automatically. Record D2 and save it.

Change only to **Localized defect near x_1 = 0.35**. Keep amplitude 0.25 and 41 points. Record D3 from both cards and charts. Locate the feature; do not infer the averaged status from the fact that a defect was inserted.

D4. Required sampling comparison

Keep the localized defect and amplitude fixed. Change **Sampling resolution → 161 points**. Record D4 and compare with D3. Note the score changes and peak location/appearance. The setting regenerates sampled synthetic data and changes numerical derivatives; it does not acquire new experimental measurements.

D5. One clean modeling run

Return to **Consistent reference / 41 points**. Click **Run modeling test →** once and wait for **Converged**. Under **Inspect → Modeling residuals**, record Δp (%) and Δy (percentage points). If convergence fails, report no modeling verdict. Save this result before changing inputs.

Use **Your conclusion** and **Export all results**. Rename files clean, canceling, localized-41 and localized-161. You may save a second clean file including the fit. The download is thermodynamic-consistency-study.json. Changing settings clears the previous fit.

Optional reopen check: expand **Study workspace · reopen and compare cases**. Under **Load studies or a comparison JSON**, select your exported file, then choose **Restore** beside its case. Check the restored controls and rerun **Run modeling test →** for a current fit. Imported results are recorded evidence, not a fresh calculation.

Finish E1/E2: an annotated residual plot, a comparison to your predictions, and a claim supported by numbers with a limitation. Take the 10-minute break at study minutes 95-105.

Lab 03: interpret what each method tests

Guide page 3 | Revisit during D5/E. This page explains methods, not the changed-case answers.

Area: a global integral

Let A be area above zero and B the positive magnitude below zero for $L=\ln(\gamma_1/\gamma_2)$. $D=100|A-B|/(A+B)$. Lab 03 integrates line segments and uses D<5% for its isothermal criterion, neglecting the excess-volume pressure term. Opposite local errors can cancel. Missing endpoints mean no full-range verdict unless an explicit extrapolation is assumed; near ideality D may be undefined.

Differential: a local relationship, then an average

$r = d[g^E/(RT)]/dx - \ln(\gamma_1/\gamma_2) + h^E/(RT^2)\ dT/dx$

Lab 03 forms g first, takes three-point finite differences at interior points and reports **100 mean |r|**, with teaching threshold **5**. Isothermal data have no temperature term. Real isobaric data need independent h^E before a corrected verdict. Averages can conceal narrow features; inspect the curve. This is not the fitted Padé/Kojima test implemented in NIST TDE.

Herington: empirical isobaric allowance

$J=150(T_{max}-T_{min})/T_{min}$, using kelvin. The lab uses D-J<10. J describes a boiling-range allowance, not measured h^E. Herington is not applicable to the required isothermal runs. Extrapolated endpoints give conditional results. The full NIST near-ideal decision tree is not implemented.

Van Ness-type: model reproduction

The isothermal fit uses two constant NRTL τ values with α=0.47 fixed. Δp=mean[100|Pcalc-Pobs|/Pobs]; Δy=mean[100|ycalc-yobs|]. Both Δp<1% and Δy<1 percentage point must hold. This teaching parameterization is not NIST TDE's five-parameter model. Small errors show reproduction, not independent predictive validation. Failure can involve model flexibility, property assumptions or data.

Build an inspectable conclusion

Label the chart axes and its case/resolution. Write a specific claim, give two relevant numerical or graphical observations, then state one limitation. Preserve unexpected results and explain a changed prediction. Do not treat resolution as new measurements or a numerical threshold as a universal physical law.

Exports save locally; they do not submit assignments. Use the actual file names/settings so the comparison can be reproduced. The generator is independent of the Lab 08 demonstration data.

Sources: NIST Herington | NIST point test | NIST Van Ness. See the Lab's Sources & scope for implementation limits.

Lab 04: three feeds and an independent energy check

Guide page 4 | Worksheet F/G, 105-155 min. Complete F and all G1 predictions before selecting named presets.

Open Lab 04: Liquid stability and LLE

1. Fix the model; compare three feeds

Set **Liquid model → Margules · 1 parameter** and **Interaction parameter A → 3**. For **Feed mole fraction z_1**, run 0.02, 0.12 and 0.50. After typing, click **Find equilibrium →**; sliders update immediately. Choose **View → Gibbs energy & common tangent**.

After predicting, the corresponding **Explore a state** presets are a shortcut. Their names disclose the homogeneous-feed labels. Check the A and z values after selection. The hypothetical model has no T/P controls; hold A fixed for this comparison.

2. Record the feed and equilibrium separately

Record **Homogeneous feed**, the final phase label, **Gibbs energy decrease / RT**, and curvature sign. For two liquids record **Coexisting composition x_1^α**, **x_1^β** and **Fraction of β liquid**. β is the component-1-rich liquid, not vapor. For one liquid, report $x_1=z_1$; coexistence fractions do not apply.

Expand **Verify the equilibrium**. Inspect chemical-potential agreement, material balance and full tangent evidence. Positive curvature is local evidence, not proof of global stability. A named unstable or metastable feed can end in a stable two-liquid equilibrium.

3. Do the independent calculations

For $z_1=0.12$ and 0.50, use the actual endpoints in $f\beta=(z_1-x^\alpha)/(x^\beta-x^\alpha)$. Compute fα and reconstruct z_1. Both fractions must lie in [0,1]. Explain what goes wrong if those two endpoints are used for $z_1=0.02$.

For G4 use $m(x)=x \ln x+(1-x)\ln(1-x)+3x(1-x)$. Evaluate m(0.12), then the weighted endpoint energy and their difference. Use natural logarithms. This includes ideal mixing; C used excess Gibbs energy only. Annotate the tangent and compare your result with the Lab energy decrease.

4. Save evidence and close the lesson

Type your reasoning in **Your explanation**, then click **Export study JSON** for each feed. Rename liquid-stability-study.json with z002, z012 or z050. Keep your prediction table, balances, energy calculation and annotated tangent. Small rounding residuals are expected if displayed endpoints are truncated.

Stop the required investigation at G4. An A sweep is optional within the block, not homework. At 155-170 min attempt H1-H4 before solutions; at 170-180 min complete the existing anonymous Session 3 form, then save all files. No attendance identity is collected.